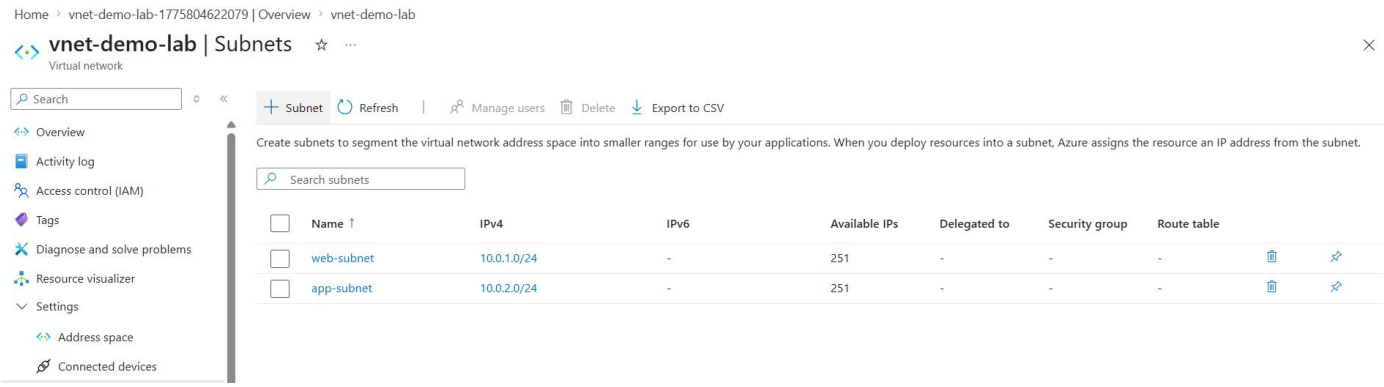
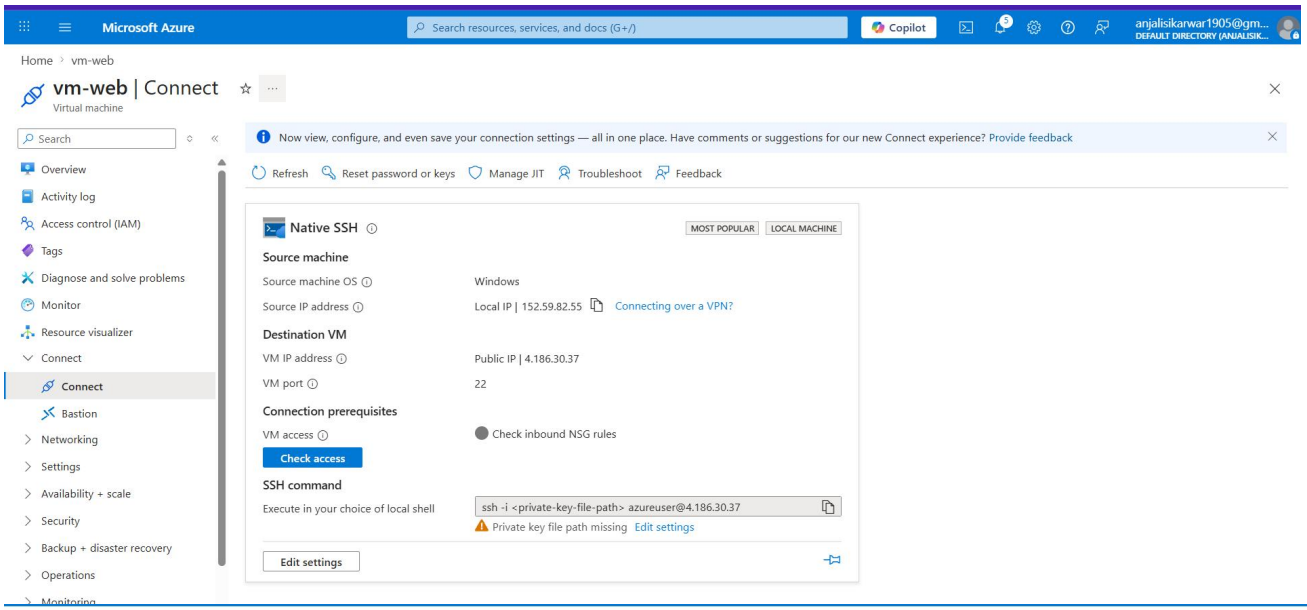
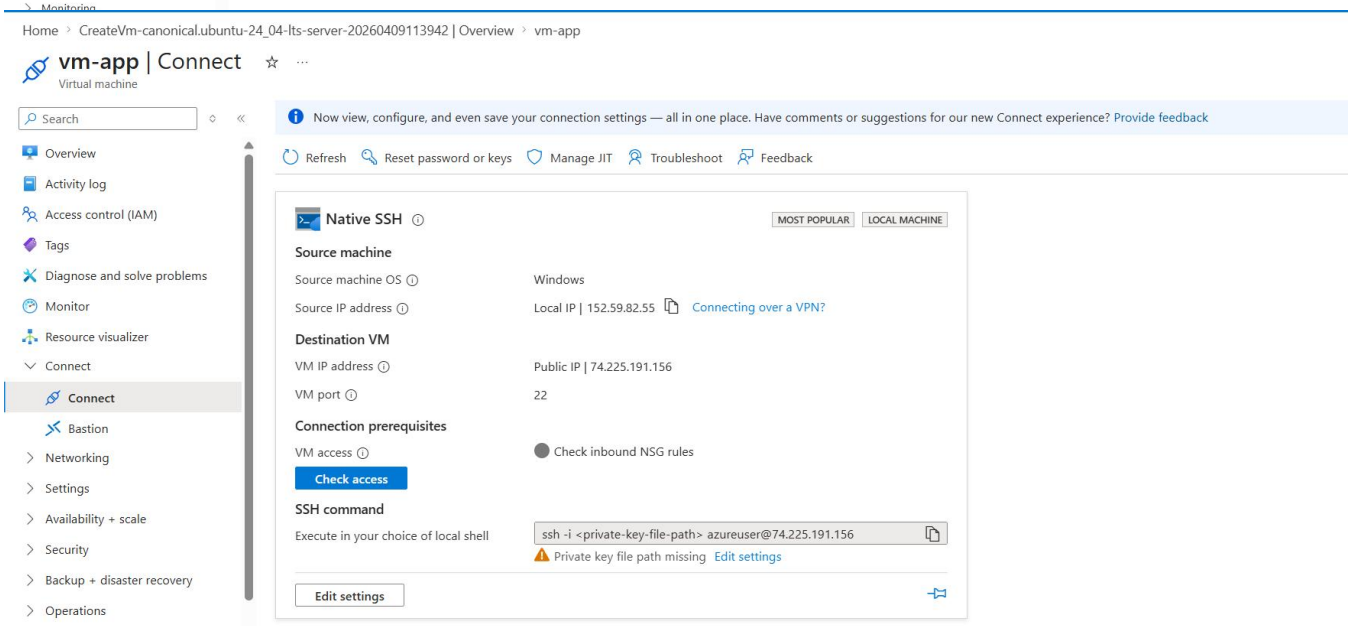
Part 1: Create Virtual Network



Part 2: Create Virtual Machines

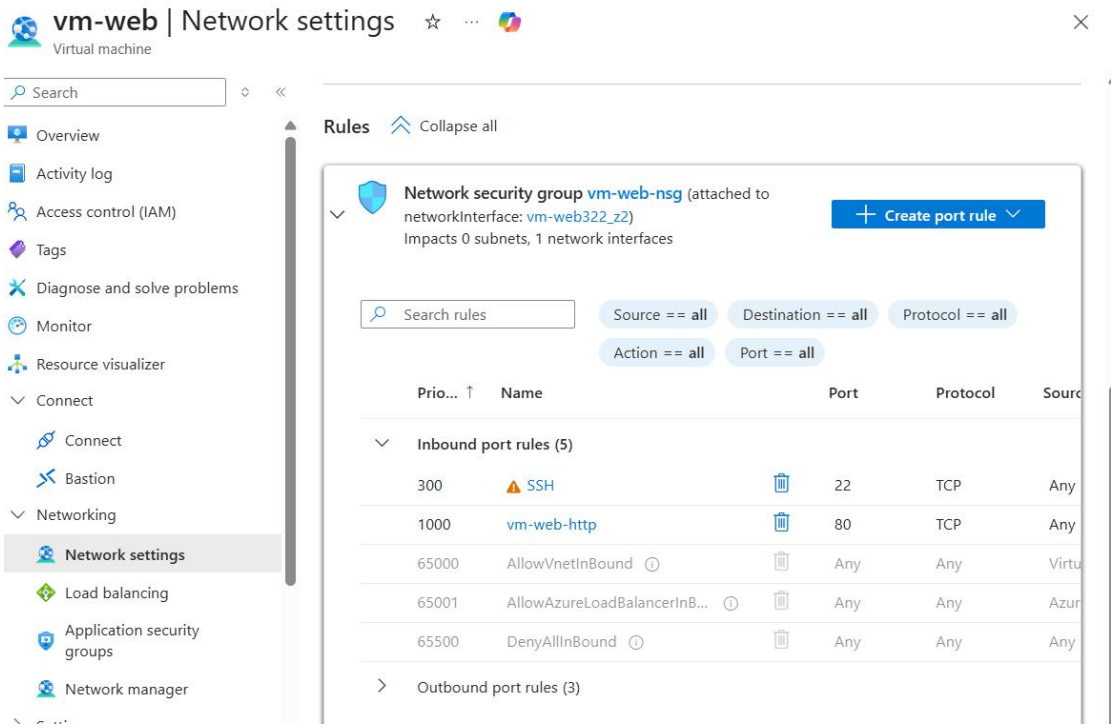





```
azureuser@vm-web:~$ sudo apt update
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
```

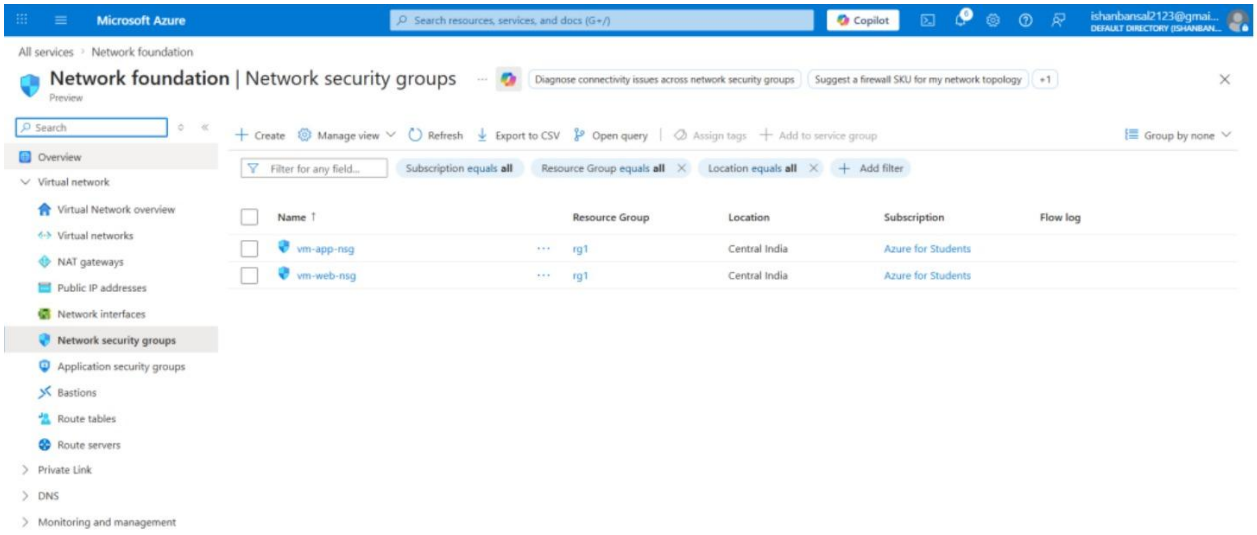
```
azureuser@vm-web:~$ sudo apt install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
```

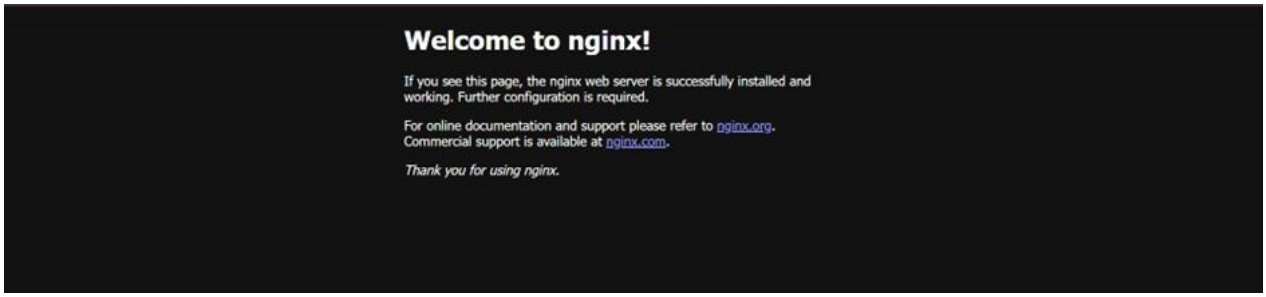
```
azureuser@vm-web:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-04-09 06:20:18 UTC; 17min ago
       Docs: man:nginx(8)
    Process: 2507 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
    Process: 2508 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 2538 (nginx)
      Tasks: 3 (limit: 9433)
     Memory: 2.4M (peak: 5.3M)
        CPU: 18ms
     CGroup: /system.slice/nginx.service
             ├─2538 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             ├─2541 "nginx: worker process"
             └─2542 "nginx: worker process"

Apr 09 06:20:18 vm-web systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Apr 09 06:20:18 vm-web systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
azureuser@vm-web:~$
```




```
azureuser@vm-web:~$ sudo ufw allow 'Nginx Full'
Rules updated
Rules updated (v6)
azureuser@vm-web:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
azureuser@vm-web:~$
```

Part 3: NSG (Network Security Group)





Part 4: Secure Network Traffic (Real Scenario)

```
azureuser@vm-app:~$ ping 10.0.2.4
PING 10.0.2.4 (10.0.2.4) 56(84) bytes of data.
64 bytes from 10.0.2.4: icmp_seq=1 ttl=64 time=0.019 ms
64 bytes from 10.0.2.4: icmp_seq=2 ttl=64 time=0.023 ms
64 bytes from 10.0.2.4: icmp_seq=3 ttl=64 time=0.018 ms
64 bytes from 10.0.2.4: icmp_seq=4 ttl=64 time=0.036 ms
64 bytes from 10.0.2.4: icmp_seq=5 ttl=64 time=0.034 ms
64 bytes from 10.0.2.4: icmp_seq=6 ttl=64 time=0.054 ms
64 bytes from 10.0.2.4: icmp_seq=7 ttl=64 time=0.055 ms
64 bytes from 10.0.2.4: icmp_seq=8 ttl=64 time=0.052 ms
64 bytes from 10.0.2.4: icmp_seq=9 ttl=64 time=0.051 ms
64 bytes from 10.0.2.4: icmp_seq=10 ttl=64 time=0.056 ms
^C
--- 10.0.2.4 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9245ms
rtt min/avg/max/mdev = 0.018/0.039/0.056/0.014 ms
azureuser@vm-app:~$
```